

# Lab 16: Deploy CockroachDB on Kubernetes with `cockroach-operator` and `kind` (70 min)

## Learning Objectives

By the end of this lab you will be able to:

- Stand up a multi-node Kubernetes cluster locally with `kind` and deploy CockroachDB via the operator
- Read the `CrdbCluster` CRD and change resources, replica count, and version through it
- Perform a rolling version upgrade and a scale-out with zero downtime
- Configure pod anti-affinity so replicas never share a failure domain
- Kill a pod and a node and watch Raft and the StatefulSet recover
- Compare the operator, the Helm chart, and manual StatefulSets — and pick one with reasons

## Prerequisites

- Docker with at least **8 GB** of memory free (a full run was measured at 7.6 GB)
- `kind` and `kubectl` — the only two tools this course installs locally, because `kind` drives the Docker daemon itself and cannot run in a container
- **Docker Desktop** (or Docker Engine) running — there is no `cockroach` binary to install

**Free the memory first.** `kind` runs four Kubernetes nodes and then three CockroachDB pods inside them. On a 12 GB machine that does not fit alongside the lab cluster, so shut it down before you start:

```
scripts/crdb down
```

Skip this and the symptom is not a clear error — pods sit in `Pending` or get OOM-killed mid-rollout, which reads like an operator bug.

```
kind version && kubectl version --client && docker info | grep -i "total memory"
```

## Setup — a Multi-Node kind Cluster

```
mkdir -p /tmp/lab16

cat > /tmp/lab16/kind-config.yaml <<'YML'
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
  - role: control-plane
  - role: worker
    labels: {topology.kubernetes.io/zone: zone-a}
  - role: worker
    labels: {topology.kubernetes.io/zone: zone-b}
  - role: worker
    labels: {topology.kubernetes.io/zone: zone-c}
YML

kind create cluster --name lab16 --config /tmp/lab16/kind-config.yaml
kubectl get nodes --show-labels | cut -c1-120
```

Three workers with distinct zone labels — that's what makes the anti-affinity in Part D real rather than decorative.

## Tasks

### Part A: Install the Operator (15 min)

#### 1. CRDs and operator:

```
kubectl apply -f https://raw.githubusercontent.com/cockroachdb/cockroach-operator/master/
install/crds.yaml
kubectl apply -f https://raw.githubusercontent.com/cockroachdb/cockroach-operator/master/
install/operator.yaml
```

2. **Wait for it — twice.** The Pod being ready is not the same as the operator's admission webhook being reachable, and applying a `CrdbCluster` in that gap fails with `failed calling webhook "mcrdbcluster.kb.io" ... connection refused`:

```
kubectl -n cockroach-operator-system rollout status deploy/cockroach-operator-manager --
timeout=180s
kubectl -n cockroach-operator-system get pods
```

```
kubectl -n cockroach-operator-system get endpoints cockroach-operator-webhook-service
```

An address listed here is necessary but not sufficient — the operator pod has no readiness probe, so its IP appears before the webhook server is listening. The only reliable test is to ask the webhook something, which is why Part B applies the cluster in a retry loop.

This is a general Kubernetes lesson, not an operator quirk: **any** operator with a mutating or validating webhook rejects resources until its Service has endpoints. If your CI applies a custom resource immediately after installing an operator, it will flake.

### 3. Look at what the CRD gives you:

```
kubectl get crd crdbclusters.crdb.cockroachlabs.com -o yaml | head -40
kubectl explain crdbcluster.spec | head -40
```

| Field                                                  | Controls                                                                 |
|--------------------------------------------------------|--------------------------------------------------------------------------|
| <code>spec.nodes</code>                                | Cluster size — change it to scale                                        |
| <code>spec.cockroachDBVersion</code>                   | Version — change it to upgrade                                           |
| <code>spec.resources</code>                            | CPU/memory requests and limits per pod                                   |
| <code>spec.dataStore.pvc</code>                        | Storage class and size per node                                          |
| <code>spec.tlsEnabled</code>                           | Operator-managed certs via cert-manager or self-signed                   |
| <code>spec.affinity / topologySpreadConstraints</code> | Failure-domain placement                                                 |
| <code>spec.additionalArgs</code>                       | Extra <code>cockroach start</code> flags (e.g. <code>--locality</code> ) |

## Part B: Deploy the Cluster (15 min)

### 1. The manifest — `/tmp/lab16/crdb.yaml` :

```

apiVersion: crdb.cockroachlabs.com/v1alpha1
kind: CrdbCluster
metadata:
  name: crdb
  namespace: default
spec:
  datastore:
    pvc:
      spec:
        accessModes: [ReadWriteOnce]
        resources:
          requests:
            storage: 2Gi
          volumeMode: Filesystem
  resources:
    requests: {cpu: "500m", memory: 1Gi}
    limits: {cpu: "1", memory: 2Gi}
  tlsEnabled: true
  cockroachDBVersion: v23.2.5
  nodes: 3
  additionalLabels:
    app: crdb
  topologySpreadConstraints:
    - maxSkew: 1
      topologyKey: topology.kubernetes.io/zone
      whenUnsatisfiable: DoNotSchedule
      labelSelector:
        matchLabels:
          app.kubernetes.io/instance: crdb

```

**Memory limits matter more here than anywhere else.** CockroachDB sizes its cache from what it believes the machine has. Under a cgroup limit, always set `--cache` and `--max-sql-memory` as *percentages* (the operator does this) and leave headroom — a pod OOM-killed mid-Raft-append is a much worse outcome than a slightly smaller cache.

## 2. Apply and watch it come up:

```

# "failed calling webhook ... connection refused" means the operator is still warming up;
# apply is idempotent, so keep trying until it is accepted.
until kubectl apply -f /tmp/lab16/crdb.yaml; do echo "webhook not ready yet – retrying";
  sleep 5; done
until kubectl get statefulset crdb >/dev/null 2>&1; do sleep 2; done # the operator
  creates it
kubectl rollout status statefulset/crdb --timeout=600s
# returns when all 3 pods are Running

```

- A client pod, then connect.** The operator generated the CA, node and root-client certificates as Secrets named after the cluster (`crdb-node`, `crdb-root`); a client pod mounts them. (The upstream example manifest hard-codes a cluster called `cockroachdb` and the latest image, so we write our own.)

```

cat > /tmp/lab16/client.yaml <<'YAML'
apiVersion: v1
kind: Pod
metadata:
  name: cockroachdb-client-secure
spec:
  serviceAccountName: crdb-sa
  nodeName: lab16-worker                # keep it off lab16-worker2, which Part D stops
  terminationGracePeriodSeconds: 0
  containers:
    - name: client
      image: cockroachdb/cockroach:v23.2.5
      command: ["sleep", "2147483648"]
      volumeMounts:
        - name: client-certs
          mountPath: /cockroach/cockroach-certs/
  volumes:
    - name: client-certs
      projected:
        defaultMode: 256
        sources:
          - secret:
              name: crdb-node
              items: [{key: ca.crt, path: ca.crt}]
          - secret:
              name: crdb-root
              items: [{key: tls.crt, path: client.root.crt}, {key: tls.key, path:
client.root.key}]
YAML
kubectl apply -f /tmp/lab16/client.yaml
kubectl wait --for=condition=Ready pod/cockroachdb-client-secure --timeout=300s
kubectl exec -it cockroachdb-client-secure -- ./cockroach sql \
  --certs-dir=/cockroach/cockroach-certs --host=crdb-public

```

```

SELECT node_id, address, is_live FROM crdb_internal.gossip_nodes;
SELECT count(*) FROM crdb_internal.ranges_no_leases;

```

#### 4. Map Kubernetes objects to CockroachDB concepts:

```
kubectl get statefulset,svc,pvc,pdb
```

| Kubernetes object                  | Role                                                                                                               |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| StatefulSet <code>crdb</code>      | Stable pod identities ( <code>crdb-0</code> , <code>crdb-1</code> , ...) — a node's identity must survive restarts |
| Headless service <code>crdb</code> | Peer discovery for <code>--join</code>                                                                             |
| Service <code>crdb-public</code>   | Client entry point                                                                                                 |
| PVC per pod                        | The store. <b>Deleting it deletes that node's data</b>                                                             |
| PodDisruptionBudget                | Stops <code>kubectl drain</code> from taking quorum with it                                                        |

#### 5. Open the DB Console:

```
kubectl port-forward svc/crdb-public 8080:8080
```

Then <https://localhost:8080> (self-signed cert — accept the warning).

## Part C: Rolling Upgrade and Scale-Out (20 min)

### 1. Generate load so you can prove "zero downtime" instead of asserting it:

```
kubectl exec -it cockroachdb-client-secure -- ./cockroach workload init kv \
  --drop 'postgresql://root@crdb-public:26257/kv?sslmode=verify-full&sslrootcert=/
cockroach/cockroach-certs/ca.crt&sslcert=/cockroach/cockroach-certs/
client.root.crt&sslkey=/cockroach/cockroach-certs/client.root.key'

kubectl exec -it cockroachdb-client-secure -- ./cockroach workload run kv \
  --duration=10m --concurrency=8 --tolerate-errors \
  'postgresql://root@crdb-public:26257/kv?sslmode=verify-full&sslrootcert=/cockroach/
cockroach-certs/ca.crt&sslcert=/cockroach/cockroach-certs/client.root.crt&sslkey=/
cockroach/cockroach-certs/client.root.key' &
```

### 2. Scale out — one field:

```
kubectl patch crdbcluster crdb --type=merge -p '{"spec":{"nodes":5}}'
kubectl get pods -w
```

### 3. Watch the data rebalance onto the new nodes:

```
SELECT node_id, count(*) AS replicas
FROM crdb_internal.ranges_no_leases, unnest(replicas) AS node_id
GROUP BY node_id ORDER BY node_id;
```

Run it a few times over a couple of minutes. The allocator moves replicas onto the new nodes without anyone telling it to.

### 4. Rolling version upgrade:

```
kubectl patch crdbcluster crdb --type=merge -p '{"spec":{"cockroachDBVersion":"v23.2.6"}}'
kubectl rollout status statefulset/crdb --timeout=600s
```

Watch the workload output during the roll. Errors should be zero (or retried, with `--tolerate-errors`), because at any moment only one node is down and quorum holds.

### 5. Finalize the upgrade — a major-version upgrade stays reversible until you finalize:

```
SHOW CLUSTER SETTING version;
-- After validating, for a major upgrade:
-- SET CLUSTER SETTING version = crdb_internal.node_executable_version();
```

### 6. Record the drill:

| Operation                 | Duration | Client errors | Notes |
|---------------------------|----------|---------------|-------|
| Scale 3 → 5               |          |               |       |
| Rebalance to steady state |          |               |       |
| Rolling upgrade           |          |               |       |

## Part D: Failure Domains and Recovery (20 min)

### 1. Confirm placement honours the zone spread:

```
kubectl get pods -o custom-columns=POD:.metadata.name,NODE:.spec.nodeName --sort-by=.spec.nodeName
kubectl get nodes -L topology.kubernetes.io/zone
```

### 2. Tell CockroachDB about the topology, not just Kubernetes. Without `--locality`, the allocator does not know two pods share a zone, and may place all three replicas of a range in one failure domain:

```
kubectl patch crdbcluster crdb --type=merge -p \
  '{"spec":{"additionalArgs":["--locality=zone=$(POD_ZONE)"]}}'
```

In a real deployment the zone comes from the downward API or an init container reading the node label. This is the single most commonly skipped step in Kubernetes deployments, and it silently defeats the whole point of running three zones.

### 3. Kill a pod:

```
kubectl delete pod crdb-2
kubectl get pods -w
```

The StatefulSet recreates it with the *same identity* and the *same PVC*, so it rejoins with its data instead of re-replicating from scratch.

### 4. Kill a whole node:

```
docker stop lab16-worker2
kubectl get nodes
```

```
SELECT node_id, is_live FROM crdb_internal.gossip_nodes;
SELECT count(*) FROM crdb_internal.ranges_no_leases; -- still answers: quorum held
```

Watch under-replicated ranges in the DB Console. After `server.time_until_store_dead` (default 5 min) the cluster starts re-replicating.

### 5. Bring it back:

```
docker start lab16-worker2
kubectl get nodes -w
```

## 6. Decommission properly — never just delete a StatefulSet pod and its PVC:

```
kubectl exec -it cockroachdb-client-secure -- ./cockroach node status \
  --certs-dir=/cockroach/cockroach-certs --host=crdb-public

# Correct order: decommission (moves replicas off), THEN scale down
kubectl exec -it cockroachdb-client-secure -- ./cockroach node decommission 5 \
  --certs-dir=/cockroach/cockroach-certs --host=crdb-public --wait=all

kubectl patch crdbcluster crdb --type=merge -p '{"spec":{"nodes":4}}'
```

**The failure mode to avoid:** scaling down first, then decommissioning. Removing the pod takes its replicas offline instantly; the cluster then has to re-replicate under duress. Decommission moves replicas off *while the node is still serving*, which is why it is slower and correct.

## Optional — If Time Allows

These parts are not required to complete the lab; they extend it by about 10 minutes. Do them if you finish early, or after class — the cluster and data from the core parts carry over.

### Part E: Operator vs Helm vs Manual (10 min)

| Approach                  | Best for                                       | Handles for you                                         | You still own                                      |
|---------------------------|------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|
| <b>cockroach-operator</b> | Most production K8s deployments                | Certs, rolling upgrades, scaling, decommission ordering | Locality flags, storage class, monitoring          |
| <b>Helm chart</b>         | Existing Helm/GitOps pipelines                 | Templating, values-driven config                        | Upgrade ordering, cert lifecycle, decommission     |
| <b>Manual StatefulSet</b> | Unusual requirements, air-gapped, full control | Nothing                                                 | Everything — including the ordering mistakes above |

#### 1. Look at the Helm alternative:

```
# helm in a container – nothing to install. The two mounts matter: without
# them `repo add` writes into a container that is deleted a moment later, and
# `show values` then reports no such repository.
mkdir -p ~/.config/helm ~/.cache/helm
alias helm='docker run --rm -v "$HOME/.config/helm:/root/.config/helm" -v "$HOME/.cache/helm:/root/.cache/helm" alpine/helm'
helm repo add cockroachdb https://charts.cockroachdb.com/
helm show values cockroachdb/cockroachdb | head -60
```

## 2. The multi-region shape (for discussion — it needs three real clusters):

- One `CrdDbCluster` per region, each with region-specific `--locality`
- Cross-cluster networking so pods can gossip (service mesh, VPC peering, or LB per region)
- `ALTER DATABASE ... SET PRIMARY REGION / ADD REGION` from Lab 7
- Survival goal set to `REGION` — which needs  $\geq 3$  regions and  $\geq 5$  replicas

## 3. Production checklist for a K8s deployment:

☐

`--locality` reflects real failure domains (zone, region)

☐

`topologySpreadConstraints` or anti-affinity prevents co-location

☐

`PodDisruptionBudget` protects quorum from voluntary disruptions

☐

Storage class is SSD-backed with `volumeBindingMode: WaitForFirstConsumer`

☐

Resource requests == limits for CPU-sensitive workloads (guaranteed QoS)

☐

Prometheus `ServiceMonitor` scraping `/_status/vars` (Lab 9)

☐

Scheduled backups to object storage, and a completed restore drill (Lab 11)

☐

`cert-manager` or the operator managing cert rotation (Lab 12)

☐

A documented decommission-then-scale-down runbook

## Cleanup

```
kind delete cluster --name lab16
```

## Lab 16 Deliverables

☐ **4-node kind cluster** with zone labels, operator installed ☐ **3-node CockroachDB** deployed via the CRD with TLS and topology spread ☐ **Scale-out** to 5 nodes with rebalancing observed ☐ **Rolling upgrade** completed under live load with client errors recorded ☐ **Pod and node failures** survived; identity and PVC reattachment understood ☐ **Correct decommission ordering** performed ☐ **Deployment comparison** with a defended choice, plus a production checklist

## Challenge Exercises

1. **cert-manager integration.** Replace the operator's self-signed certs with cert-manager issued certs and rotate them without downtime (Lab 12's SIGHUP applies here too).
2. **Prove the anti-affinity.** Remove `topologySpreadConstraints`, redeploy, and check whether replicas of the same range land in one zone. How would you detect this in production before an outage does it for you?
3. **Break decommissioning on purpose.** Scale from 5 → 3 by patching `nodes` *without* decommissioning first. What does `SHOW RANGES` report? How do you recover?
4. **Backups from inside K8s.** Add a `CronJob` that runs `BACKUP INTO 's3://...'` and alerts on failure. Where do the credentials live, and why not in the manifest?

## Reference

| Command                                                                            | Purpose                                      |
|------------------------------------------------------------------------------------|----------------------------------------------|
| <code>kind create cluster --config</code>                                          | Multi-node local Kubernetes                  |
| <code>kubectl apply -f ../operator.yaml</code>                                     | Install the CockroachDB operator             |
| <code>kubectl explain crdbcluster.spec</code>                                      | Discover CRD fields                          |
| <code>kubectl patch crdbcluster crdb --type=merge -p '{"spec":{"nodes":N}}'</code> | Scale                                        |
| <code>kubectl patch ... '{"spec":{"cockroachDBVersion":"vX.Y.Z"}}'</code>          | Rolling upgrade                              |
| <code>kubectl rollout status statefulset/crdb</code>                               | Watch the roll                               |
| <code>cockroach node decommission N --wait=all</code>                              | Move replicas off before removing a node     |
| <code>topologySpreadConstraints</code>                                             | Keep replicas out of the same failure domain |
| <code>--locality=zone=...</code>                                                   | Tell CockroachDB about the topology          |